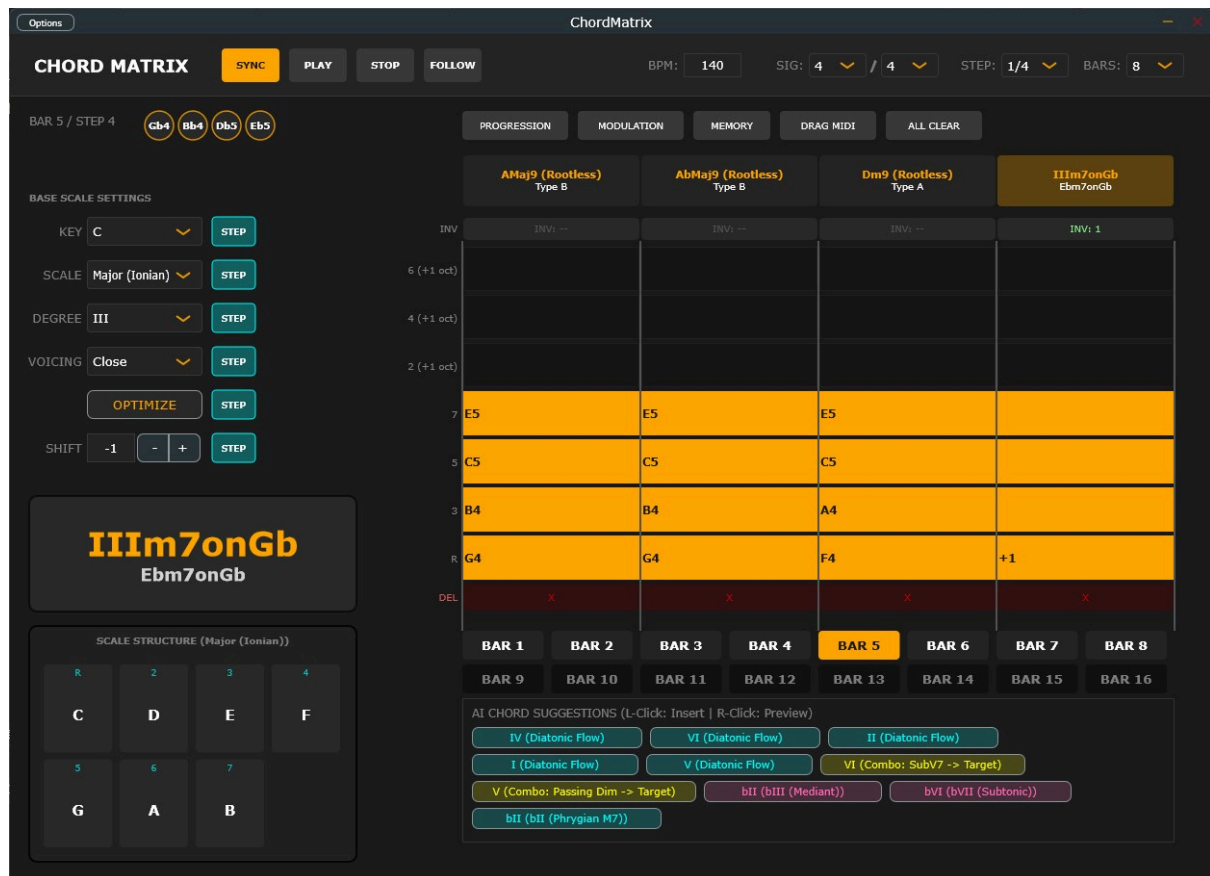


ChordMatrix User's Manual

This manual provides a detailed explanation of all the functions of the next-generation MIDI Chord Sequencer "ChordMatrix," as well as the musical theoretical and mathematical mechanisms behind them.



1. Basic Concepts and Degree Notation

ChordMatrix is based on "Degree Notation" (Degree Name), which does not rely on specific note names (like C or G). This allows the role (function) of the chord to be maintained even when changing keys, making it applicable to any song.

1.1 How to Read Degree Notation

The Roman numerals displayed on the UI indicate which scale degree the root note is based on within the selected scale.

- **Uppercase (I, II, IV, V...):** Major chord systems.
- **Lowercase (i, ii, iv, v...):** Minor chord systems.
- **Added Symbols (#IV, bVII...):** Roots that are chromatically altered from the standard notes of the scale.
- **Special (dim, m7b5...):** Specifies the nature (quality) of the chord.

1.2 Why Degree Notation?

Based on the mathematical system of jazz harmony, chords are processed as a set of "pitch classes" (numbers 0 to 11). By adopting degree notation, users can focus on the musical narrative (function) of "Tonic" (resolution) and "Dominant" (tension) during composition.

2. Scale Database (Total 55 Types) Details

ChordMatrix is equipped with 55 types of scales covering everything from Western music to ethnic and contemporary music. These are designed with "DSP Safety" and are reflected in the voicing calculations in real time.

2.1 Diatonic / Church Modes

The seven most basic scales.

- **Major (Ionian)**: Bright sound. The foundation of everything.
- **Natural Minor (Aeolian)**: The standard minor. A sound of melancholy.
- **Dorian**: Frequently used in jazz and funk. A slightly brighter minor.
- **Lydian**: The "#4" note creates a floating and mysterious feeling. Ideal for film scores.
- **Mixolydian**: Basic scale for blues and rock. A slightly muddled major.

2.2 Melodic Minor Derivatives (Modern Jazz)

The most important group for creating the "emotional" sound of modern jazz and Neo-Soul.

- **Altered (Super Locrian)**: Creates intense tension with the dominant chord (V7).
- **Lydian Dominant**: A stylish, urban sound used in chords like "IV7."

2.3 Symmetric and Special Scales

- **Diminished (HW / WH)**: Strong tension, used in horror, suspense, and as passing tones in jazz.
- **Whole Tone**: All whole-step intervals. An atonal sound, like being in a dream.
- **Bebop Scales**: 8-note structure for smoothly connecting fast phrases in jazz.

2.4 Ethnic/World

- **Yonanuki (Yonanuki)**: The soul of Japanese popular songs and Enka.
- **Ryukyu (Ryukyu)**: The distinctive, open sound of Okinawa.
- **Hungarian Minor**: A passionate minor sound, like Gypsy music.

3. Chord Presets (Total 90 Types) Utilization

Chord progressions used by professionals are categorized into 14 groups.

3.1 Representative Categories

- **US: Trap & Neo-Soul**: Contemporary R&B with heavy low frequencies, utilizing 9th and 11th chords extensively.
- **J-POP: Royal Road (Oudō Kōshin)**: Progressions centered on IV - V - iii - vi, touching the heartstrings of Japanese listeners.

- **J-POP: Marunouchi (Marunouchi Sadistic Progression):** A stylish progression based on IV - III - vi - I.
- **Pro: Masterpiece (16 bars):** Professional-level, long progressions that depict an epic narrative.

3.2 Preset Intelligence

Every preset is linked to a "Recommended Scale." The moment a preset is loaded, the Scale setting in the inspector is automatically optimized, maximizing the accuracy of the AI's next chord suggestion.

4. The Two Major Intelligences of the AI Engine

The core of ChordMatrix lies not just in the placement of chords, but in how they are "connected."

4.1 Optimization via Viterbi Algorithm (Optimize)

Instead of just looking at "movement from A to B," this algorithm uses dynamic programming to calculate the "path that minimizes finger travel distance throughout the entire phrase."

- **5 Performance Personalities (AltIndex):**
 1. **Best Balance:** The mathematical shortest path.
 2. **Melody Lead:** The top voice (top note) creates a beautiful melody.
 3. **Bass Lead:** Emphasizes the independence of the bass line.
 4. **Compact:** A modern sound with notes clustered toward the center.
 5. **Alternate Ending:** Aims for a resolution with an element of surprise.

4.2 Context-Adaptive Modulation

When a target key is specified, the AI analyzes the preceding chord progression.

- **Pivot Chord (Common Chord):** Finds a chord belonging to both keys for a smooth bridge.
- **Tritone Sub (Substitute Dominant):** Lands dramatically from the dominant chord a half-step above.
- **Passing Diminished:** Creates a line where the bass smoothly ascends chromatically.
- **Neo-Riemannian:** A dramatic modulation that flips the color while retaining common tones.

5. UI Operation Guide

5.1 Scope Setting (STEP / BAR / ALL)

Allows switching the range of application with one button.

- **STEP (Cyan):** Applies to only the selected cell.
- **BAR (Orange):** Applies to the entire selected bar. Maintains consistency with preceding and succeeding bars.

- **ALL (HotPink):** Processes all bars in a batch. Integrates the voice leading for the entire song.

5.2 AI Suggestion

Clicking an empty cell prompts the AI to suggest 10 candidates, considering both "statistical certainty" and "musical surprise."

- **Combo Suggestion:** Prioritizes combos such as "ii - V" aimed at the target, not just individual chords.
- **Lock Function:** When a Lock pin is inserted into a favorite chord, the AI re-calculates the surrounding chords using that chord as a "fixed point."

6. Technical Hints

- **Automatic Avoidance of Avoid Notes:** The Viterbi engine mathematically detects dissonances within the selected scale and automatically alters tensions (9th/13th, etc.) to altered forms (b9, b13, etc.) for resolution.
- **Voicing Mode:** Selecting "Drop 2" lowers the second note from the top by an octave, resulting in a beautiful, less clustered sound.

ChordMatrix is the bridge between your inspiration and music theory. Enjoy free creation based on the mathematically-backed "correctness."